A kinetic-impact deflection model

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Estimate a velocity change and explain the value of lead time.

SUCCESS CRITERIA - I can...

1. State the simplified collision model.
2. Connect delivered momentum to target velocity change.
3. Explain why real deflection includes ejecta and orbital effects.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

01 Warm-up: bring back an earlier idea

Two equal 3 kg carts approach at 5 m/s in opposite directions and stick. Find their common final velocity and explain.

Marzano check - circle: ① ② ③

02 Worked example: explain the model

Reference: $\Delta v \approx p_{projectile}/M_{target}$; shift ≈ Δv × time.
Toy model: much heavier target; straight-line shift.

Worked reasoning: In a fictional model, a 2 kg projectile at 30 m/s sticks to a stationary 10 kg target. Final speed is 60/12=5 m/s. The system includes both objects.

Your turn: Draw the before-and-after system boundary and momentum arrows.

Before	After

Label mass, velocity and momentum.

Marzano check - circle: ① ② ③

03 Read and interpret the model

After a brief sideways push, a probe has a constant additional sideways velocity. Compare the offset after a short drift and a long drift. Use the model to connect impulse, velocity change and later position. Assume no further sideways force.

Impulse J → Change Δv → Offset Δx

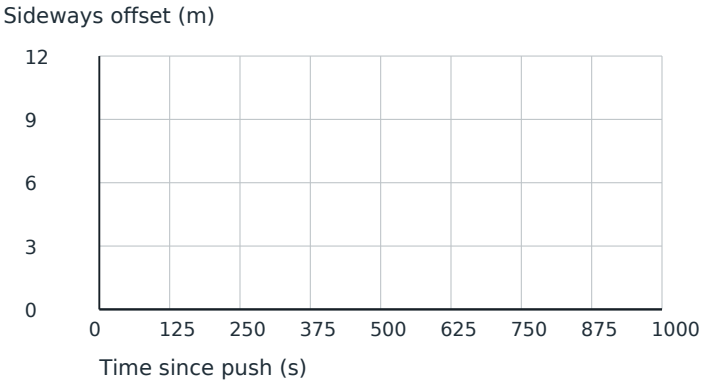
Marzano check - circle: ① ② ③

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04 Apply the physics

A simulated probe receives a brief push that adds 0.01 m/s sideways to its velocity. It then drifts without further sideways force. Complete the graph and find its sideways offset after 100 s and 1000 s.



Marzano check - circle: 1 2 3

05 Explain your reasoning

Why should the no-ejecta sticking calculation not be reported as the exact measured result of a real mission?

Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

An asteroid’s period is 6 years. Explain why that number alone cannot establish an Earth encounter. Name two additional observations needed.
Organize the timing evidence before drawing a conclusion.

Object	Arrival time / window	Comparison
A	___	___
B	___	___

Marzano check - circle: 1 2 3

07 Exit check

At the same small Δv, compare shift with twice as much lead time. State one model limit. Organize the known quantities on the before-and-after model.

Before	After

Label objects and signed velocities.

Marzano check - circle: 1 2 3

08 STRETCH: test the model

Return to the worked example in problem 2. Keep projectile momentum 60 kg·m/s but double the stationary target mass from 10 to 20 kg. Calculate the new final speed and explain the change. Organize the known quantities on the before-and-after model.

Before	After

Label objects and signed velocities.

Marzano check - circle: 1 2 3